

Amirhossein Rajabpour

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Headline

Bridging ML research and engineering • Machine Learning, Reinforcement Learning, Generalization

Education

- **University of Alberta** 2023 – 2025
M.Sc. in Computing Science, advised by [Levi Lelis](#) & [Sandra Zilles](#)
GPA: 3.8/4
Research Topics: Reinforcement Learning, Deep Learning, Program Synthesis
Edmonton, AB, Canada
- **Amirkabir University of Technology (Tehran Polytechnic)** 2018 – 2023
B.Sc. in Computer Engineering (Major: AI, Minor: Computer Networks)
GPA (last two years): 3.96/4
Tehran, Iran

Publications

- *Common Benchmarks Undervalue the Generalization Power of Programmatic Policies.*
– **A. Rajabpour**, K. Aghakasiri, S. Zilles, L. Lelis. (RLC 2025 Programmatic RL Workshop). [\[Preprint\]](#) [\[Code\]](#) [\[Page\]](#)

Technical Skills

- **Programming:** Python, C, Java, Kotlin, Octave
- **ML/AI:** PyTorch, TensorFlow, Keras, scikit-learn, fastai, OpenCV
- **Data/DB:** MySQL, PostgreSQL, MongoDB
- **Web:** Django, Flask, HTML, CSS
- **Tools/OS:** Git, Docker, Jira, Selenium, XAMPP, Linux, Windows, macOS

Professional and Research Experience

- **AI Researcher**, Huawei Technologies Canada Oct 2025 – Present
– Working on efficient inference of Large language models on Smartphones (LiteRT Framework).
- **Research Assistant**, University of Alberta May 2024 – Jul 2025
– Refined neural policies (RL agents) through targeted architectural and training adjustments, enabling them to match or outperform programmatic policies in generalization on key benchmarks.
– Proposed new benchmarks highlighting strengths of programmatic policies (e.g., memory-based reasoning).
– Built end-to-end training/evaluation pipelines (Python, Bash, PyTorch, TensorFlow).
– Used **Large Language Models** to generate/assess interpretable policies where RL underperforms.
– Work accepted at [RLC 2025 Programmatic RL workshop](#).
- **Technical Advisor**, Alberta Machine Intelligence Institute (Amii) Sep 2024 – Oct 2024
– 1-month project on advanced AI/ML methods to enhance real-time intrusion detection and anomaly detection in high-speed network storage infrastructures.
– Focused on threat-detection accuracy and scalability in multi-cloud settings.
- **Machine Learning Engineer**, R&D Dept., Crouse PJS Co. Oct 2021 – Jan 2022
– Designed lightweight industrial ML for detecting malfunctioning LEDs: localization (OpenCV), fuzzy C-means clustering of light pixels, luminance/wavelength extraction, and per-color regression modeling.
- **DevOps Engineer Intern**, Graph Co. Nov 2020 – Apr 2021
– Hands-on experience with Docker and cloud-native tooling.

Additional Research Projects

- **Individual Study Course Project**, University of Alberta Jan 2023 – Apr 2024
– Implemented hill climbing (with exploration) and a genetic algorithm to improve “[Unveiling Options with Neural Decomposition](#)”. Supervisor: Levi Lelis.
- **Reinforcement Learning 1 Course Project**, University of Alberta Sep 2023 – Dec 2023
– Compared generalization of expert (concentrates resources on a single task) vs. generalist (splits resources across tasks) agents (PPO, DQN, APPO) across MiniGrid and budget regimes. [\[Project Report & Code\]](#)
- **Bachelor’s Thesis: Face Aging Platform**, Amirkabir University Mar 2023 – May 2023
– Developed a face-aging platform using generative models (e.g., CycleGAN, reversible models). Supervisor: Mohammad Rahmati. [\[Code\]](#)

- Integrated multiple pretrained models with distinct dependencies into a unified system using Docker Compose, deploying each in separate containers and orchestrating communication across them.
- **Research Assistant: Portfolio Allocation with RL & GNNs** , Amirkabir University Feb 2022 – Jun 2022
 - Implemented a graph convolutional network over time-series for asset allocation. Supervisor: Hamed Farbeh.

Teaching Assistant

- **University of Alberta**
 - CMPUT 365: Introduction to Reinforcement Learning | Instructor: Marlos Machado Fall 2025
 - CMPUT 366: Search & Planning in AI | Instructor: Levi Lelis Fall 2024, Winter 2025
 - CMPUT 291: Intro to File & Database Management | Instructors: Davood Rafiei, Arash Nobari Fall 2023, Winter 2024
- **Amirkabir University of Technology**
 - Principles of AI | Instructor: Mahdi Javanmardi Fall 2022
 - Cloud Computing | Instructor: S. Ahmad Javadi Fall 2022
 - Internet of Things | Instructor: Siavash Khorsandi Fall 2022
 - Algorithm Design | Instructors: A. Bagheri, S. Shirali-Shahreza Winter 2022, Fall 2021, Winter 2021

Honors and Awards

- Master's admissions: Fully funded offers from UAlberta (CS, ECE, Radiology & Diagnostic Imaging) & Western Ontario (CS) 2023
- Bachelor's Scholarship: 4-year scholarship, Amirkabir University of Technology 2019–2023
- Nationwide Mathematics University Entrance Exam (Iran): Top 1% among 140,000+ applicants 2018

Graduate Courses

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| • Reinforcement Learning 1 (Marlos Machado) | • Modelling Strategic Behavior (James Wright) |
| • Reinforcement Learning 2 (Rich Sutton) | • Machine Learning (Lili Mou) |

Language Skills

- Persian: Native
- English: TOEFL iBT 109 (R 26, L 28, S 26, W 29)
- German: Professional Working Proficiency (B2)